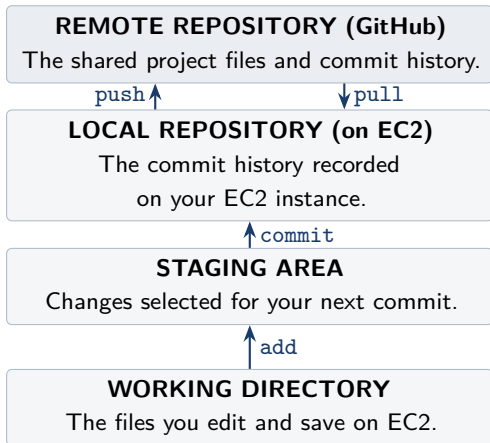


Git & GitHub: The Minimum You Need

FRE6883 – Recitation 1 – Deck 3



Git records versions.

GitHub hosts a remote repository.

Where is “local”?

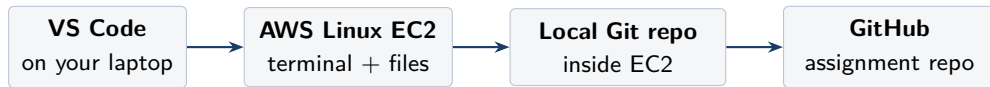
In this course, Git runs on **AWS Linux EC2**. VS Code connects to EC2 so you can edit those files.

The repeated path

Save → add → commit → push.

Pull brings remote updates into your local repository and updates your working files.

Start in the right place



```
git clone <url>
```

Copies an existing remote repository and its history into a new local folder.

```
git status
```

Shows which files are modified, staged, or untracked in the current repository.

For your assignment

After accepting the Classroom 50 invitation, copy your own GitHub repository URL. Clone once into ~/FRE6883 on EC2. Open the cloned folder in VS Code; run the remaining commands inside that folder.

Replace <url> with your URL, without brackets. Already cloned? Open the existing folder.

Save, select, record

```
git add <file>
```

Stages the current saved changes in the named file for the next commit.

```
git add .
```

Stages all non-ignored changes under the current folder, including additions and deletions.

```
git commit -m "message"
```

Records the staged changes as a new local commit with a short description.

Two habits that prevent surprises

Save before adding. If you edit the file again, save it and add it again.

Before `git add .`, check `git status`. Make sure every listed file belongs in your commit.

A commit records the staged version; it stays on EC2 until you push.

Send your work; retrieve updates

```
git push
```

Sends your local commits to the remote repository on GitHub.

```
git pull
```

Retrieves remote updates and incorporates them into your local repository and working files.

When starting work

Run `git status`, then `git pull` when your working directory is clean.

Have local edits? Save and commit your work first.

When sharing work

Save, add, commit, then `git push`.

Open your assignment repository on GitHub and verify the latest commit appears.

If Git stops with an error, read the message and ask for help.

Inspect before you act

```
git diff
```

Shows saved, unstaged changes to files Git already tracks.

```
git log
```

Shows the commit history leading to your current local version.

Three different questions

```
git status
```

 Which files need attention?

```
git diff
```

 What lines have I changed since staging?

```
git log
```

 What work has already been committed?

`git diff` omits staged changes and new untracked files. Use `git status` to see their state.

If the output opens in a pager, press `q` to return to the terminal.

Scenario 1: get your code onto GitHub

You modified `Option01.cpp` and want your changes to appear on GitHub. **What commands should you run?**

Assume you are in the correct EC2 repository and have saved the file.

Think for 20 seconds; compare your sequence with a neighbor.

Scenario 1: get your code onto GitHub

You modified `Option01.cpp` and want your changes to appear on GitHub. **What commands should you run?**

Assume you are in the correct EC2 repository and have saved the file.

Think for 20 seconds; compare your sequence with a neighbor.

Expected sequence

```
git status
git add Option01.cpp
git commit -m "Implement CRR pricer"
git push
```

After a successful push, refresh your GitHub repository to check the commit.

Scenario 2: the remote repository changed

You changed your local files. Your teammate or instructor has also updated the **same remote repository** on GitHub.

Which command retrieves those remote updates?

Think for 15 seconds, then answer.

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Which command retrieves those remote updates?

Think for 15 seconds, then answer.

Answer

```
git pull
```

First save your files, check `git status`, and add and commit your local work. Then pull the remote updates.

If Git cannot complete the operation, read its message and ask for help.

Scenario 3: what changed, and what was recorded?

You saved edits to the tracked file `Option01.cpp`, but have not staged them.

1. Which command shows the changed lines?
2. Which command shows earlier commits and their messages?

Think for 15 seconds, then answer.

Scenario 3: what changed, and what was recorded?

You saved edits to the tracked file `Option01.cpp`, but have not staged them.

1. Which command shows the changed lines?
2. Which command shows earlier commits and their messages?

Think for 15 seconds, then answer.

Answers, in order

```
git diff
```

```
git log
```

`git diff` inspects your unstaged edits; `git log` inspects recorded commits.

Common mistakes

- **Forgetting to save the file.**

Save in VS Code before running `git add`.

- **Forgetting `git add`.**

Stage the changes you want in the next commit.

- **Committing but forgetting to push.**

Run `git push`, then verify the commit on GitHub.

- **Editing in the wrong folder or repository.**

Check the EC2 connection, open folder, and terminal location.

- **Running commands blindly.**

Read `git status` and Git's output before the next step.

FRE6883: VS Code → AWS Linux EC2 → local Git repository → GitHub / Classroom 50

Command	Purpose
<code>git clone <url></code>	Copy a remote repository and its history locally; do this once.
<code>git status</code>	Check modified, staged, and untracked files.
<code>git add <file></code>	Stage the saved changes in one file.
<code>git add .</code>	Stage all non-ignored changes under the current folder.
<code>git commit -m "message"</code>	Record the staged changes as a local commit.
<code>git push</code>	Send local commits to GitHub.
<code>git pull</code>	Retrieve remote updates and apply them locally.
<code>git log</code>	Read the local commit history.
<code>git diff</code>	Inspect saved, unstaged changes to tracked files.

In your EC2 repo: save → `git status` → add → commit → push → verify on GitHub.

To start: check status; pull with a clean working directory. Save and commit local edits first.